EXISTENCE OF THE LOCALIZED CHERN CLASS AS A BIVARIANT CLASS ON A CLOSED SUBSCHEME, WITH THE PUSHFORWARD IDENTITY ONTO THE ZERO FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0F9K (sources/stacks-chow/pinned-original-source.tex line 10204).

Allowed standard prerequisites (editorial).

- The Gysin-at-infinity bivariant class C and its defining property. Same chapter, Lemma labelled lemma-gysin-at-infty (source line 10026), invoked at proof lines 10236 and 10267. Provides C in A^0(W_infinity -> X) with i_infinity,*(C cap alpha) = i_{0,*}alpha, which is the ingredient that moves the construction from W_infinity down to X.
- The local construction of the class c'_p(Q|_E) on a closed subscheme where the module is zero or of small rank. Same chapter, Lemma labelled lemma-silly (source line 9695), invoked at proof lines 10237-10239. Supplies c'_p(Q|_E) in A^p(E -> W_infinity), applicable because of assumption (A2); the assembly of the three classes is inside the counted span.
- Proper pushforward of bivariant classes and the comparison lemma for the restricted class. Same chapter, Lemma labelled lemma-push-proper-bivariant (source line 6167) at proof line 10247 and Lemma labelled lemma-silly-silly (source line 9792) at proof line 10265. Supply that the composite defining c'_p(Q) is again a bivariant class and the comparison with c_p(Q|_{W_infinity}).
- Chern classes of a perfect object and the base situation (S, delta). Same chapter, Section labelled section-pre-derived (source line 9042) and Situation labelled situation-setup (source line 1372), invoked in the statement. Fix assumption (A0) and the ambient operational Chow groups.

Source reference. Original definition 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof.

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It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ESY: sources/stacks-chow/pinned-original-source.tex, source lines 9042–9042. Same-chapter pointer into the author environment lemma at source line 9042; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9J: sources/stacks-chow/pinned-original-source.tex, source lines 10026–10026. Same-chapter pointer into the author environment lemma at source line 10026; invoked at proof line 10237 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9H: sources/stacks-chow/pinned-original-source.tex, source lines 9695–9695. Same-chapter pointer into the author environment lemma at source line 9695; invoked at proof line 10239 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EPK: sources/stacks-chow/pinned-original-source.tex, source lines 6167–6167. Same-chapter pointer into the author environment lemma at source line 6167; invoked at proof line 10247 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9I: sources/stacks-chow/pinned-original-source.tex, source lines 9792–9792. Same-chapter pointer into the author environment lemma at source line 9792; invoked at proof line 10265 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $Z \subset X$ be a closed subscheme. Let*

$$b : W \longrightarrow \mathbf{P}^1_X$$

be a proper morphism of schemes. Let $Q \in D(\mathcal{O}_W)$ be a perfect object. Denote $W_\infty \subset W$ the inverse image of the divisor $D_\infty \subset \mathbf{P}^1_X$ with complement $\mathbf{A}^1_X$. We assume

- (A0) *Chern classes of Q are defined (Section 0ESY),*
- (A1) *b is an isomorphism over $\mathbf{A}^1_X$,*
- (A2) *there exists a closed subscheme $T \subset W_\infty$ containing all points of W_∞ lying over $X \setminus Z$ such that $Q|_T$ is zero, resp. isomorphic to a finite locally free $\mathcal{O}_T$-module of rank $< p$ sitting in cohomological degree 0.*

Then there exists a canonical bivariant class

$$P'_p(Q), \text{ resp. } c'_p(Q) \in A^p(Z \to X)$$

with $(Z \to X)_* \circ P'_p(Q) = P_p(Q|_{X \times \{0\}})$, resp. $(Z \to X)_* \circ c'_p(Q) = c_p(Q|_{X \times \{0\}})$.

Proof. Denote $E \subset W_\infty$ the inverse image of Z. Then $W_\infty = T \cup E$ and b induces a proper morphism $E \to Z$. Denote $C \in A^0(W_\infty \to X)$ the bivariant class constructed in Lemma 0F9J. Denote $P'_p(Q|_E)$, resp. $c'_p(Q|_E)$ in $A^p(E \to W_\infty)$ the bivariant class constructed in Lemma 0F9H. This makes sense because $(Q|_E)|_{T \cap E}$ is zero, resp. isomorphic to a finite locally free $\mathcal{O}_{E \cap T}$-module of rank $< p$ sitting in cohomological degree 0 by assumption (A2). Then we define

$$P'_p(Q) = (E \to Z)_* \circ P'_p(Q|_E) \circ C, \text{ resp. } c'_p(Q) = (E \to Z)_* \circ c'_p(Q|_E) \circ C$$

This is a bivariant class, see Lemma 0EPK. Since $E \to Z \to X$ is equal to $E \to W_\infty \to W \to X$ we see that

$$\begin{aligned}
(Z \to X)_* \circ c'_p(Q) &= (W \to X)_* \circ i_{\infty,*} \circ (E \to W_\infty)_* \circ c'_p(Q|_E) \circ C \\
&= (W \to X)_* \circ i_{\infty,*} \circ c_p(Q|_{W_\infty}) \circ C \\
&= (W \to X)_* \circ c_p(Q) \circ i_{\infty,*} \circ C \\
&= (W \to X)_* \circ c_p(Q) \circ i_{0,*} \\
&= (W \to X)_* \circ i_{0,*} \circ c_p(Q|_{X \times \{0\}}) \\
&= c_p(Q|_{X \times \{0\}})
\end{aligned}$$

The second equality holds by Lemma 0F9I. The third equality because $c_p(Q)$ is a bivariant class. The fourth equality by Lemma 0F9J. The fifth equality because $c_p(Q)$ is a bivariant class. The final equality because $(W_0 \to W) \circ (W \to X)$ is the identity on X if we identify W_0 with X as we've done above. The exact same sequence of equations works to prove the property for $P'_p(Q)$. $\square$